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Fields of Concentration:

Primary Field: International Trade
Secondary Field: Economic Development

Qualifying Examinations Completed:

2023: International Trade (*with distinction*), Economic Development (*with distinction*)

Dissertation Title: *Essays on International Trade, Technology, and Firms*

Committee:

Professor Peter Schott (Chair)
Professor Amit Khandelwal
Professor Samuel Kortum
Professor Teresa Fort

Education:

Ph.D., Economics, Yale University, 2027 (expected)
M.Phil., Economics, Yale University, 2024
M.A., Economics, Yale University, 2023
B.A., Mathematics and Economics (*with highest distinction*), University of Virginia, 2020

Fellowships, Honors and Awards:

Ryoichi Sasakawa Young Leaders Fellowship (SYLFF), 2023-2024
Cowles Foundation Fellowship, 2021-2026
Doctoral Fellowship, Yale University, 2021-2026
Best Undergraduate Paper Award, International Atlantic Economic Society, 2020

Teaching Experience:

Spring 2026, Teaching Assistant to Prof. Peter Schott, Decision Making with Data (MBA),
Yale University School of Management

Fall 2025, Teaching Assistant to Prof. Teresa Delgado, Long-Term Fiscal Sustainability
(undergraduate), Yale College
Summer 2024, Co-Director, Herb Scarf Summer Research Program in Economics
(undergraduate), Yale University

Research Experience:

Research Assistant to Prof. Peter Schott and Prof. Teresa Fort, Yale University, 2023-2025
Research Assistant to Prof. Paul Goldsmith-Pinkham, Yale University, Summer 2022

Work Experience:

Data Scientist, 2nd Order Solutions, 2020-2021

Publications:

“[Selective Entry in Highway Procurement Auctions](#)” (2020), *Atlantic Economic Journal*,
48, 519–533.

Working Papers:

“Offshoring and Knowledge Spillovers”, *Job Market Paper*

“Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly
Released Software” with Tra Nguyen

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References:

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Dissertation Abstract

Offshoring and Knowledge Spillovers (*Job Market Paper*)

Since the 1990s, U.S. manufacturers have moved production of goods they once made at home to China. Offshoring lowered their labor costs, but in making the goods, their affiliates and suppliers in China also gained the know-how required for production. This production knowledge may have diffused to nearby producers, strengthening Chinese companies that export to the United States in competition with U.S. manufacturers. While import competition from China has received substantial attention, less is known about such offshoring-induced knowledge spillovers and what they may have contributed to China's exports to the United States.

This paper studies offshoring as a channel of knowledge diffusion from the U.S. to China and quantifies its implications for U.S. welfare. I construct the first measure of U.S. offshoring built on observed domestic production, covering the universe of U.S. manufacturers in 1997. Combining the Census of Manufactures with the Longitudinal Firm Trade Transactions Database, I match U.S. firms' imports over 1997–2017 to their domestic production at the industry level, and classify a firm's imports in an industry and year as *offshored production* if the importing firm also manufactured in that industry in the U.S. in 1997. Using the buyer-supplier links in the import records, I examine whether offshoring relationships differ from non-offshoring trade relationships in ways consistent with knowledge transfer.

Does this knowledge reach Chinese exporters that never receive offshoring orders? Spillovers should be largest within a city and industry: production knowledge is more applicable within an industry than across industries, and diffusion is most likely between nearby producers. I therefore infer diffusion from how the export performance of these exporters responds when any of their neighbors first receives an offshoring order. I exploit variation in the timing of the first offshoring order across city-industry cells through a staggered difference-in-differences design. Identification relies on treated and not-yet-treated cells following similar outcome trends absent offshoring, conditional on size-specific trends and sector-city-year fixed effects. I estimate the effect from five years before to five years after the first order, tracing the dynamic path to show whether and how quickly spillovers materialize.

To evaluate the welfare consequences, I then develop a quantitative model of offshoring and knowledge diffusion in which U.S. firms choose whether to offshore production to China. Offshoring lowers labor costs but also transfers production knowledge that diffuses to nearby Chinese producers. Firms internalize neither the tougher foreign competition that diffusion creates for other U.S. firms nor the lower prices it brings to U.S. consumers, so their private offshoring decisions may differ from the U.S. social planner's choice. I quantify this wedge and evaluate corrective policies: a Pigouvian tax or subsidy on offshoring, tariffs, and trade agreements with intellectual property and trade secret provisions.

(Results are omitted because Census disclosure review has not yet been completed.)

Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly Released Software, with Tra Nguyen

Workforce aging may affect economic growth not only through lower aggregate labor supply, but also by delaying technology adoption. This paper provides causal evidence on how local workforce age structure affects firms' adoption of newly released software. Using Lightcast job postings, we measure employer demand for 15 frontier software technologies released between 2012 and 2016, each paired with a legacy technology used for similar tasks. We instrument local age structure, defined as the share of college-educated workers aged 50 to 69, with a Bartik-style measure based on the historical age composition of commuting zones and national survival rates. We find that within five years of release, a one-standard-deviation increase in the older college-educated share lowers postings requiring frontier software skills by 15–19 percent of the pre-release demand for the paired legacy technology. This gap is larger when the transition is more costly, as measured by an LLM-based index of retraining cost for each legacy–frontier pair. A calibrated model attributes the gap to finite horizons: older workers have less time to recoup retraining costs, making late-career skill investment less attractive. Counterfactual exercises show that aging driven by weak inflows of young workers, as under low fertility or low migration, slows technology diffusion, while raising the retirement age partly offsets the slowdown by strengthening older workers' incentives to retrain.